

Name: _____
Instructor: _____

CHE 205 – Chemical Process Principles

Spring 2018

Test #2. Closed book – Closed notes.

- Begin **each** numbered problem **on a NEW sheet of paper**,
- Show all your work, and **BOX your final answers**.
- Only work written on the **FRONT** of the paper will be graded.
- Write all your answers on the paper provided – **DO NOT** record any answers on the test paper itself.
- The numbers in parentheses next to each part of each problem are point values for that part.

HONOR PLEDGE: “I have neither given nor received unauthorized aid on this test. I will not discuss the exam with anyone until after 5 pm on Friday, March 16, 2018.”

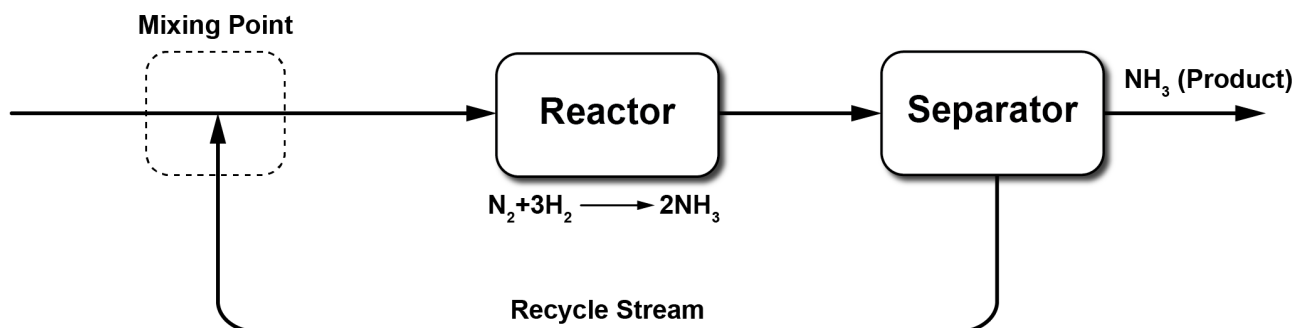
Your name (printed)

Your signature

(For recording scores by the graders)

Q. 1	
Q. 2	
Q. 3	
Q. 4	
Total	

1. (51 pts total) Ammonia (NH_3) is formed from N_2 and H_2 in a continuous steady-state reaction-separation process. The fresh feed to the process has a flow rate of 224 L(STP)/s and contains 25.0 mol% N_2 and the balance H_2 . This stream is joined by a recycle stream which contains 24.8 mol% N_2 , 74.4 mol% H_2 , and balance NH_3 . The combined stream flows into the reactor. The reactor product stream, which contains all three species, passes through a separation unit in which 95.2 % of the ammonia (molar flow rate) is collected as the product stream leaving the overall process. The gas stream leaving the separation unit is recycled back into the system.

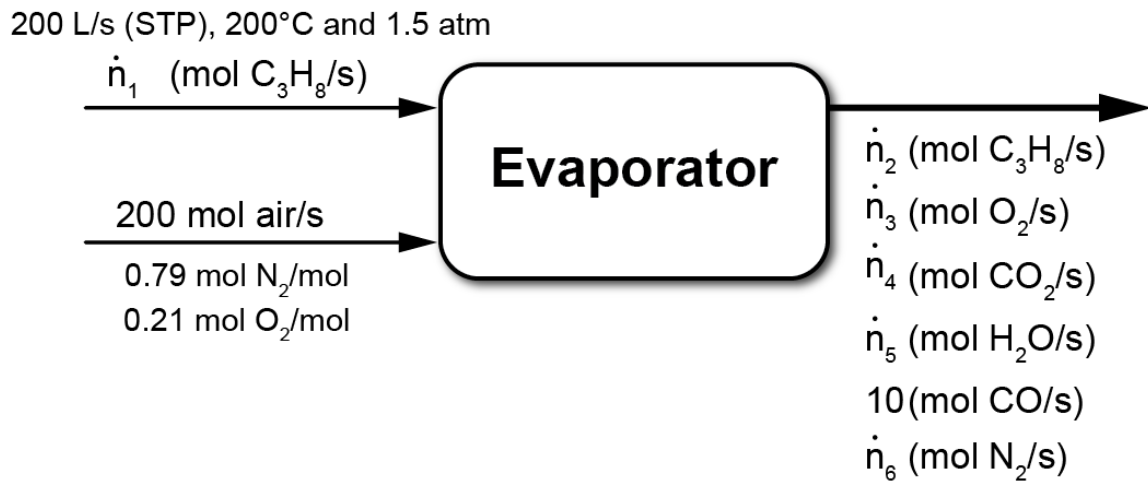


- (a) (10 pts) Re-draw the given flow chart, completely label all process streams, and identify the **basis** of calculation.
- (b) (4 pts) In terms of your labeled variables, write expressions for the overall and single-pass conversions of N_2 . **DO NOT perform any calculations.**
- (c) (4 pts) Do the degree-of-freedom (DOF) analysis for the overall process using atomic balances.
- Hint: Note that you are only allowed one atomic balance for the overall process.*
- (d) (3 pts) Explain why only 1 atomic balance is allowed for the overall process.
- (e) (4 pts) Calculate the production rate of ammonia (mol/s)
- (f) (15 pts) Do additional DOF analyses on the three subsystems shown on the flow chart and determine an efficient order in which the subsystems should be analyzed to determine all remaining unknown variables on the chart.
- Hint: Remember to use molecular balances for non-reactive subsystems.*
- (g) (8 pts) Based on your analysis in **Part (f)**, write the complete set of equations you would solve to calculate the recycle stream flow rate (mol/s). You should have n equations and n unknowns. Do NOT perform any calculations. DO NOT write more than you need to solve for the recycle flow rate.
- (h) (3 pts) Suppose the feed to the process contained trace amounts of argon, an inert gas. Why and how would the process have to be modified?

2. (12 pts total) A gas separation unit is used to separate oxygen from nitrogen in the air. Prior to separation, air is compressed to 10 atm in a multi-stage compressor and then cooled to -136.6°C . The air flowrate is 200 SCMM (Standard Cubic Meter per Minute). Calculate the volumetric flow rate of air in cubic meters per minute (m^3/min) after it has been compressed and cooled using:

- (a) (3 pts) The ideal gas equation of state
- (b) (7 pts) Kay's rule and the compressibility factor equation of state
- (c) (2 pts) Is the ideal gas equation of state a good approximation under these conditions? Briefly justify your answer.

3. (22 pts total) A flammable gas (C_3H_8) at 200°C and 1.5 atm flows into a furnace at 200 L (STP)/s and undergoes incomplete combustion with air. Air is fed to the furnace at 200 mol/s. The products emerge from the furnace containing 10 mol/s CO. The conversion of C_3H_8 is 90%. The flowsheet is provided below:



- (a) (4 pts) Write out and balance the relevant reaction(s) for this system.
- (b) (6 pts) Do a degree-of-freedom analysis around the furnace using the **atomic balance** approach.
- (c) (2 pts) Can you assume that the flammable gas behaves as an ideal gas? Justify your answer.
- (d) (10 pts) Write out (but do NOT solve) all the equations that would be required to solve for the unknowns on your flow diagram (use atomic balance approach)

4. (10 pts total) Short answer:

- a) (5 pts) Methane flows through a pipe at 44.8 SCMH. What is the molar flow rate in mol/hr at 100°C and 2 atm?
- b) (5 pts) Ethylene(C_2H_4) reacts with oxygen (O_2) to form ethylene oxide (C_2H_4O) in a reactor fed with 50 lb-mol of C_2H_4 and 50 lb-mol of O_2 . Which reactant is in excess? What is the percentage excess of the excess reactant?

Extra Information:

- General Balance Equation: **Input + Generation – Output – Consumption = Accumulation**

- Equation for the area of a circle with diameter D : $Area = \pi \frac{D^2}{4}$

- Volume of a sphere with a circle of R : $Volume = \frac{4}{3}\pi R^3$

- Gravity: $g = 9.81 \text{ m/s}^2 = 32.174 \text{ ft/s}^2$

- Gravitational conversion constant: $g_c = \frac{1 \text{ N}}{1 \text{ kg}\cdot\text{m/s}^2}$, $g_c = \frac{1 \text{ lb}_f}{32.174 \text{ lb}_m\cdot\text{ft/s}^2}$

- Pressure exerted by a column of fluid: $P = \rho gh$

- Absolute and gauge pressure: $P_{abs} = P_g + P_{atm}$

- Specific gravity: $\rho = SG \cdot \rho_{ref}$

- Reference density of water: $1.000 \text{ g/ml} = 1000 \text{ kg/m}^3 = 62.427 \text{ lb}_m/\text{ft}^3$

- Temperature Conversions:

$$[^\circ\text{F}] = 1.8[^\circ\text{C}] + 32$$

$$[\text{K}] = [^\circ\text{C}] + 273.15$$

$$[^\circ\text{R}] = 1.8([^\circ\text{C}] + 273.15)$$

$$[^\circ\text{R}] = [^\circ\text{F}] + 459.67$$

- Standard temperature and pressure (STP) for gases: 0°C , 1 atm

- Molar volume of a gas at STP: 22.414 L/mol

- Ideal gas equation of state: **$PV = nRT$**

- The gas constant, **R** :

$$= 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} = 0.08206 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K}) = 62.36 \text{ L}\cdot\text{mmHg}/(\text{mol}\cdot\text{K})$$

$$= 0.7302 \text{ ft}^3\cdot\text{atm}/(\text{lb-mol}\cdot^\circ\text{R}) = 10.73 \text{ ft}^3\cdot\text{psia}/(\text{lb-mol}\cdot^\circ\text{R})$$

- RT/P** cut-offs for assuming ideal gas behavior: **5 L/mol** (diatomic gases), **20 L/mol** (other gases)

- SRK equation of state:

$$\omega_{\text{water}} = 0.344$$

$$P = \frac{RT}{\hat{V} - b} - \frac{aa}{\hat{V}(\hat{V} + b)} \quad a = 0.42747 \frac{(RT_c)^2}{P_c} \quad m = 0.48508 + 1.55171\omega - 0.1561\omega^2$$
$$b = 0.08664 \frac{RT_c}{P_c} \quad T_r = T/T_c \quad \alpha = \left[1 + m(1 - \sqrt{T_r})\right]^2$$

- Pseudocritical T_c , P_c for mixtures: $T'_c = \sum y_i T_{c,i}$, $P'_c = \sum y_i P_{c,i}$

- Reduced temperature and pressure: $T_r = T/T_c$, $P_r = P/P_c$

- General extent of reaction balance: $n_{out} = n_{in} + \sum v_i \xi_j$

- O₂: MW = 32.00 g/mol, $T_c = 154.4 \text{ K}$, $P_c = 49.7 \text{ atm}$

- N₂: MW = 28.02 g/mol, $T_c = 126.2 \text{ K}$, $P_c = 33.5 \text{ atm}$

- Partial pressure: $P_i = y_i P$

- Extent of reaction equation for a batch reactor:

$$n_i = n_{i,0} + \sum_i \nu_i \xi$$

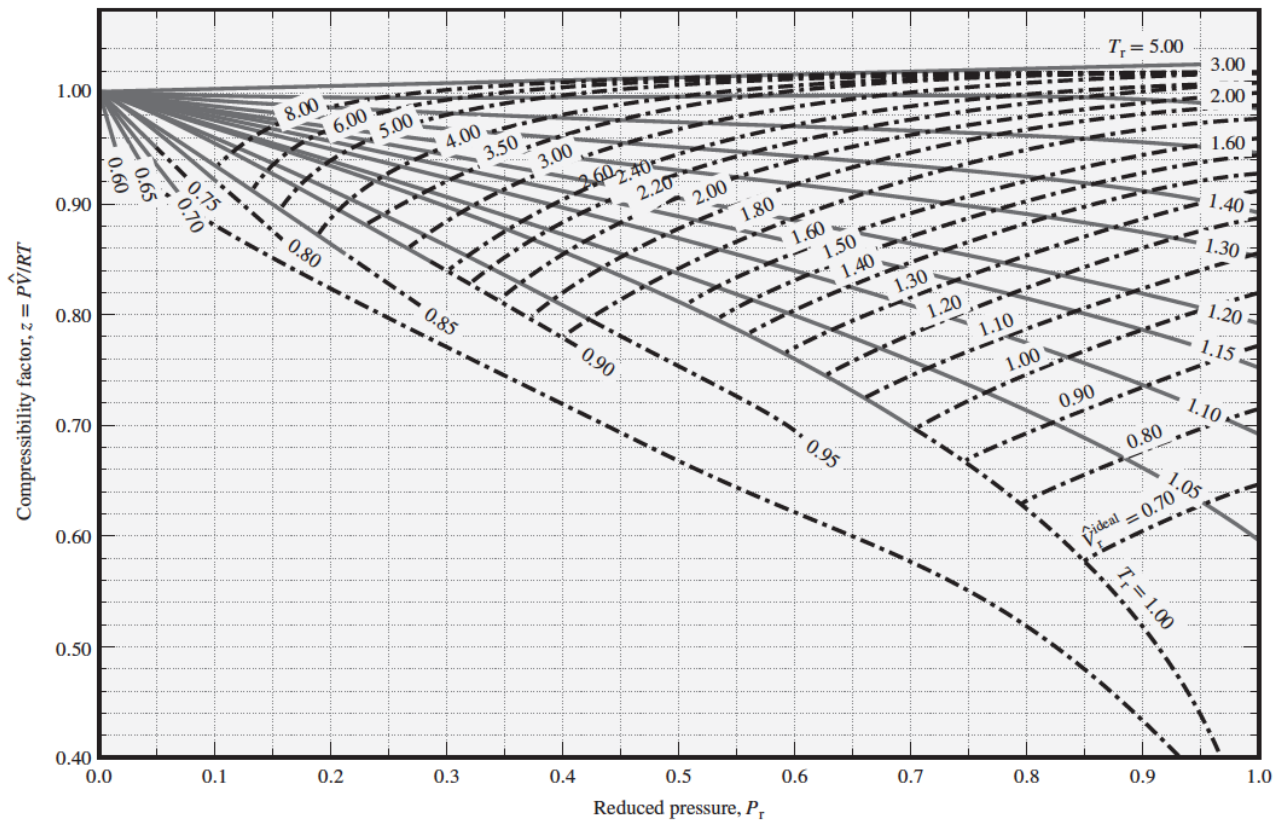
- Selectivity = moles of desired product / moles of undesired product
- Yield = moles of desired product / moles that would have formed if there were no side reactions and the reaction went to completion

- Fractional excess = $\frac{n_{i,0} - n_{i,theoretical}}{n_{i,theoretical}}$

- Fractional conversion = $\frac{n_{i,initial} - n_{i,final}}{n_{i,initial}}$

- Fractional conversion = moles reacted / moles fed

- Compressibility factor equation of state: $P\hat{V} = zRT$



Quantity	Equivalent Values
Mass	$1 \text{ kg} = 1000 \text{ g} = 0.001 \text{ metric ton (tonne)} = 2.20462 \text{ lb}_m = 35.27392 \text{ oz}$ $1 \text{ lb}_m = 16 \text{ oz} = 5 \times 10^{-4} \text{ ton} = 453.593 \text{ g} = 0.453593 \text{ kg}$
Length	$1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm} = 10^6 \text{ microns } (\mu\text{m}) = 10^{10} \text{ angstroms } (\text{\AA})$ $= 39.37 \text{ in} = 3.2808 \text{ ft} = 1.0936 \text{ yd} = 0.0006214 \text{ mile}$ $1 \text{ ft} = 12 \text{ in} = 1/3 \text{ yd} = 0.3048 \text{ m} = 30.48 \text{ cm}$
Volume	$1 \text{ m}^3 = 1000 \text{ L} = 10^6 \text{ cm}^3 = 10^6 \text{ mL} = 35.3145 \text{ ft}^3$ $= 219.97 \text{ imperial gallons} = 264.17 \text{ gal} = 1056.68 \text{ qt}$ $1 \text{ ft}^3 = 1728 \text{ in}^3 = 7.4805 \text{ gal} = 29.922 \text{ qt} = 0.028317 \text{ m}^3 = 28.317 \text{ L}$
Density	$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3 = 62.43 \text{ lb}_m/\text{ft}^3$ $= \text{density of liquid water at } 4^\circ\text{C (reference for specific gravities)}$
Force	$1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2 = 10^5 \text{ dynes} = 10^5 \text{ g}\cdot\text{cm/s}^2 = 0.22481 \text{ lb}_f$ $1 \text{ lb}_f = 32.174 \text{ lb}_m\cdot\text{ft/s}^2 = 4.4482 \text{ N} = 4.4482 \times 10^5 \text{ dynes}$
Pressure	$1 \text{ atm} = 1.01325 \times 10^5 \text{ N/m}^2 (\text{Pa}) = 101.325 \text{ kPa} = 1.01325 \text{ bar}$ $= 1.01325 \times 10^6 \text{ dynes/cm}^2 = 14.696 \text{ lb}_f/\text{in}^2 (\text{psi})$ $= 760 \text{ mm Hg at } 0^\circ\text{C (torr)} = 10.333 \text{ m H}_2\text{O(l) at } 4^\circ\text{C}$ $= 29.921 \text{ inches Hg at } 0^\circ\text{C} = 406.8 \text{ inches H}_2\text{O(l) at } 4^\circ\text{C}$
Energy	$1 \text{ J} = 1 \text{ N}\cdot\text{m} = 10^7 \text{ ergs} = 10^7 \text{ dyne}\cdot\text{cm} = 1 \text{ kg}\cdot\text{m}^2/\text{s}^2$ $= 2.778 \times 10^{-7} \text{ kW}\cdot\text{h} = 0.23901 \text{ cal} = 0.23901 \times 10^{-3} \text{ kcal (food calorie)}$ $= 0.7376 \text{ ft}\cdot\text{lb}_f = 9.486 \times 10^{-4} \text{ Btu}$
Power	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ N}\cdot\text{m/s} = 0.23901 \text{ cal/s} = 0.7376 \text{ ft}\cdot\text{lb}_f/\text{s}$ $= 9.486 \times 10^{-4} \text{ Btu/s} = 1.341 \times 10^{-3} \text{ hp}$

Example: The factor to convert grams to lb_m is $\left(\frac{2.20462 \text{ lb}_m}{1000 \text{ g}}\right)$ or $\left(\frac{1 \text{ lb}_m}{453.593 \text{ g}}\right)$.